```
#Date: 2026-09-13
#By: Divyam Sood
#Compare two numbers

.data
n1: .asciiz "Enter first number:"
n2: .asciiz "Enter second number:"
equall: .asciiz "both numbers are equal"
state: .asciiz " is greater than "

.globl main
.text

main:
    la $a0, n1
    li $v0, 4
    syscall
    li $v0, 6
    syscall
    mov.s $f1, $f0
    la $a0, n2
    li $v0, 4
    syscall
    li $v0, 6
    syscall
    #$f0 has second number

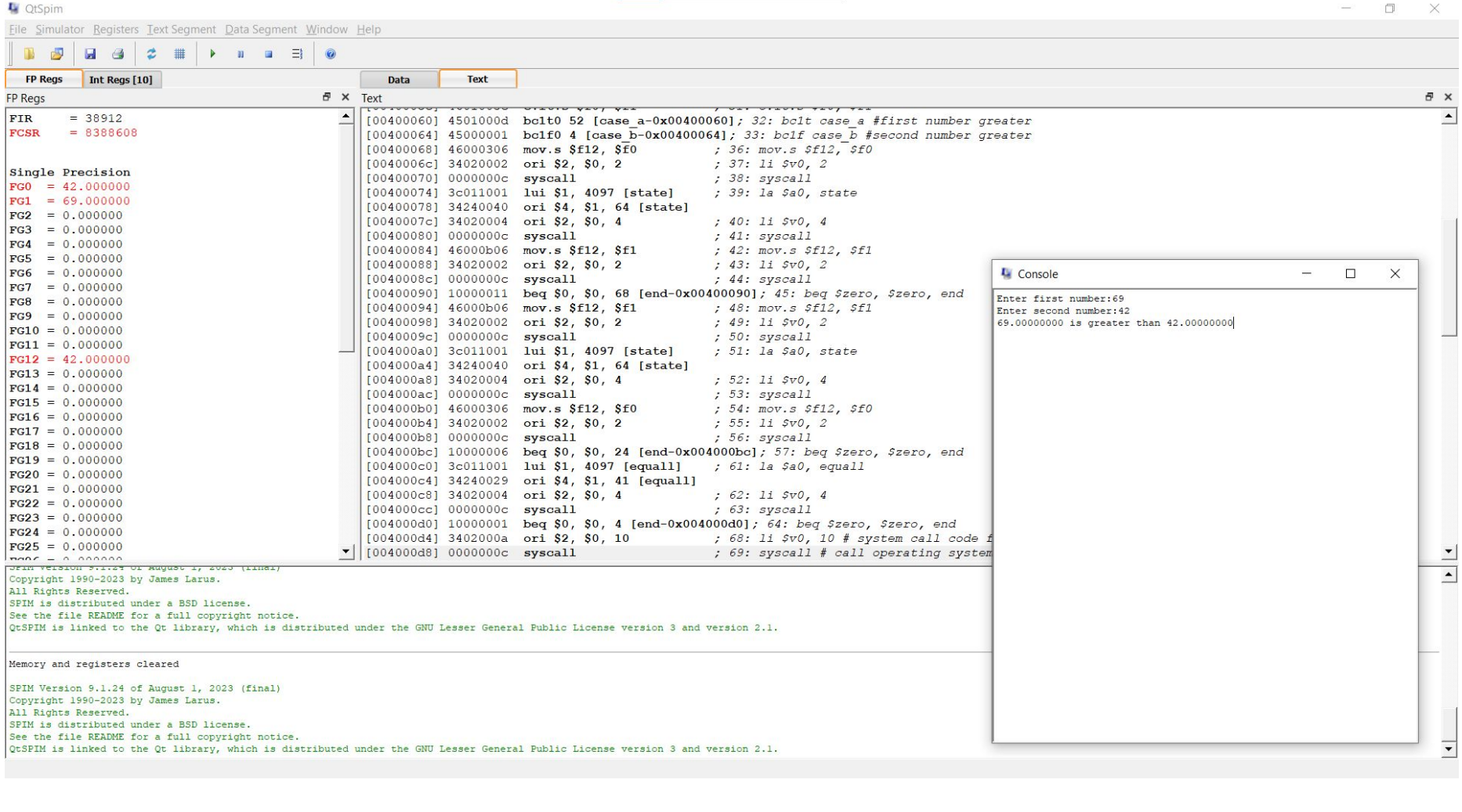
    #Check equality
    c.eq.s $f0, $f1
    bc1t equal
    c.lt.s $f0, $f1
    bc1t case_a #first number greater
    bc1f case_b #second number greater

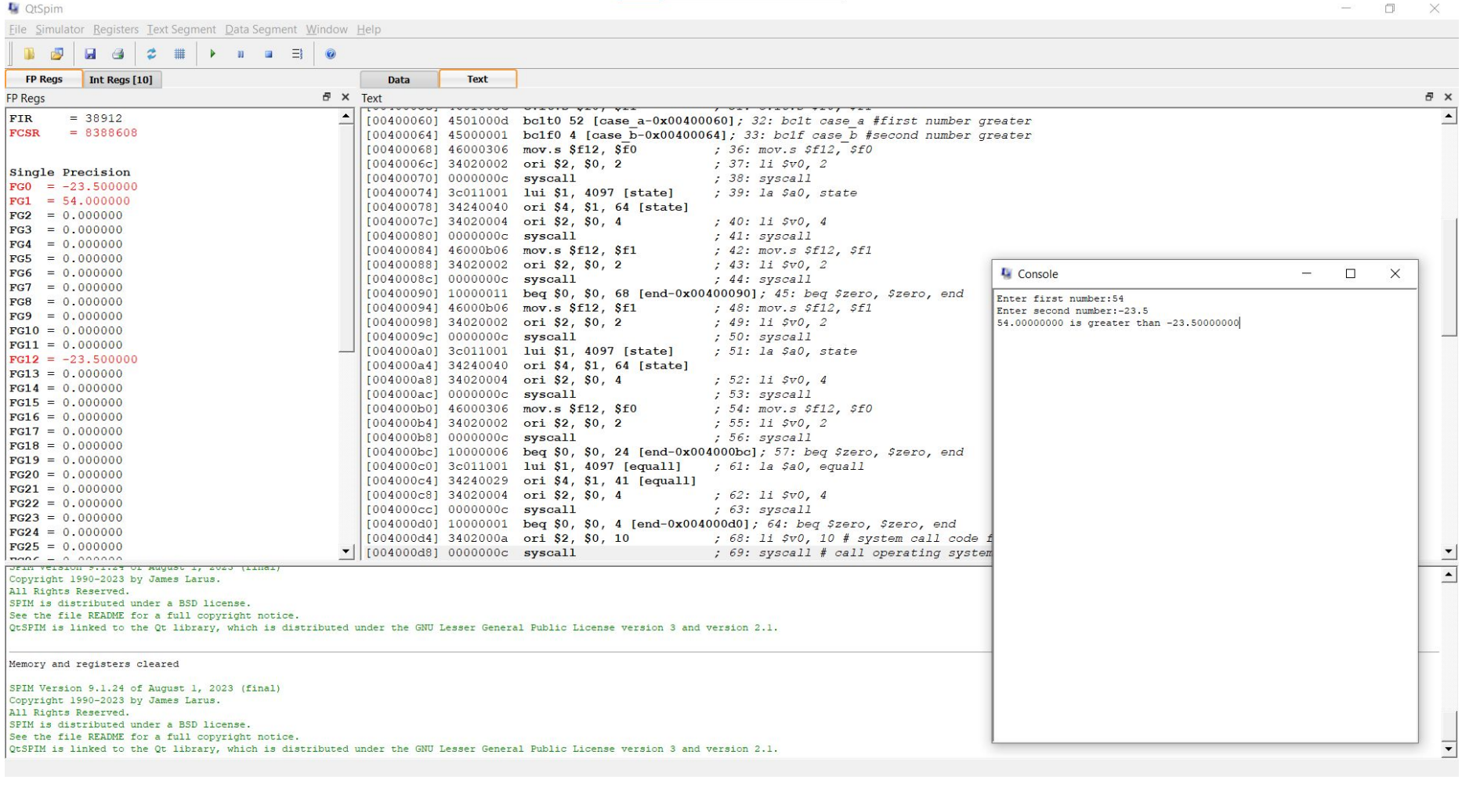
case_b:
    mov.s $f12, $f0
    li $v0, 2
    syscall
    la $a0, state
    li $v0, 4
    syscall
    mov.s $f12, $f1
    li $v0, 2
    syscall
    beq $zero, $zero, end

case_a:
    mov.s $f12, $f1
    li $v0, 2
    syscall
    la $a0, state
    li $v0, 4
    syscall
    mov.s $f12, $f0
    li $v0, 2
    syscall
    beq $zero, $zero, end


equal:
    la $a0, equall
    li $v0, 4
    syscall
    beq $zero, $zero, end

end:
    # exit program
    li  $v0, 10         # system call code for exit = 10
    syscall             # call operating system
```


QtSpim
File Simulator Registers Text Segment Data Segment Window Help
FP Regs
Int Regs [10]
Data
Text
FP Regs
FIR = 38912
FCSR = 8388608
Single Precision
FG0 = 42.000000
FG1 = 69.000000
FG2 = 0.000000
FG3 = 0.000000
FG4 = 0.000000
FG5 = 0.000000
FG6 = 0.000000
FG7 = 0.000000
FG8 = 0.000000
FG9 = 0.000000
FG10 = 0.000000
FG11 = 0.000000
FG12 = 42.000000
FG13 = 0.000000
FG14 = 0.000000
FG15 = 0.000000
FG16 = 0.000000
FG17 = 0.000000
FG18 = 0.000000
FG19 = 0.000000
FG20 = 0.000000
FG21 = 0.000000
FG22 = 0.000000
FG23 = 0.000000
FG24 = 0.000000
FG25 = 0.000000
Text
[00400060] 4501000d bc1t0 52 [case_a-0x00400060]; 32: bc1t case_a #first number greater
[00400064] 45000001 bc1f0 4 [case_b-0x00400064]; 33: bc1f case_b #second number greater
[00400068] 46000306 mov.s $f12, $f0 ; 36: mov.s $f12, $f0
[0040006c] 34020002 ori $2, $0, 2 ; 37: li $v0, 2
[00400070] 0000000c syscall ; 38: syscall
[00400074] 3c011001 lui $1, 4097 [state] ; 39: la $a0, state
[00400078] 34240040 ori $4, $1, 64 [state]
[0040007c] 34020004 ori $2, $0, 4 ; 40: li $v0, 4
[00400080] 0000000c syscall ; 41: syscall
[00400084] 46000b06 mov.s $f12, $f1 ; 42: mov.s $f12, $f1
[00400088] 34020002 ori $2, $0, 2 ; 43: li $v0, 2
[0040008c] 0000000c syscall ; 44: syscall
[00400090] 10000011 beq $0, $0, 68 [end-0x00400090]; 45: beq $zero, $zero, end
[00400094] 46000b06 mov.s $f12, $f1 ; 48: mov.s $f12, $f1
[00400098] 34020002 ori $2, $0, 2 ; 49: li $v0, 2
[0040009c] 0000000c syscall ; 50: syscall
[004000a0] 3c011001 lui $1, 4097 [state] ; 51: la $a0, state
[004000a4] 34240040 ori $4, $1, 64 [state]
[004000a8] 34020004 ori $2, $0, 4 ; 52: li $v0, 4
[004000ac] 0000000c syscall ; 53: syscall
[004000b0] 46000306 mov.s $f12, $f0 ; 54: mov.s $f12, $f0
[004000b4] 34020002 ori $2, $0, 2 ; 55: li $v0, 2
[004000b8] 0000000c syscall ; 56: syscall
[004000bc] 10000006 beq $0, $0, 24 [end-0x004000bc]; 57: beq $zero, $zero, end
[004000c0] 3c011001 lui $1, 4097 [equall] ; 61: la $a0, equall
[004000c4] 34240029 ori $4, $1, 41 [equall]
[004000c8] 34020004 ori $2, $0, 4 ; 62: li $v0, 4
[004000cc] 0000000c syscall ; 63: syscall
[004000d0] 10000001 beq $0, $0, 4 [end-0x004000d0]; 64: beq $zero, $zero, end
[004000d4] 3402000a ori $2, $0, 10 ; 68: li $v0, 10 # system call code f
[004000d8] 0000000c syscall ; 69: syscall # call operating system
Console
Enter first number:69
Enter second number:42
69.00000000 is greater than 42.00000000
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Memory and registers cleared
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QtSpim
File Simulator Registers Text Segment Data Segment Window Help
FP Regs
Int Regs [10]
Data
Text
FP Regs
FIR = 38912
FCSR = 8388608
Single Precision
FG0 = -23.500000
FG1 = 54.000000
FG2 = 0.000000
FG3 = 0.000000
FG4 = 0.000000
FG5 = 0.000000
FG6 = 0.000000
FG7 = 0.000000
FG8 = 0.000000
FG9 = 0.000000
FG10 = 0.000000
FG11 = 0.000000
FG12 = -23.500000
FG13 = 0.000000
FG14 = 0.000000
FG15 = 0.000000
FG16 = 0.000000
FG17 = 0.000000
FG18 = 0.000000
FG19 = 0.000000
FG20 = 0.000000
FG21 = 0.000000
FG22 = 0.000000
FG23 = 0.000000
FG24 = 0.000000
FG25 = 0.000000
Text
[00400060] 4501000d bc1t0 52 [case_a-0x00400060]; 32: bc1t case_a #first number greater
[00400064] 45000001 bc1f0 4 [case_b-0x00400064]; 33: bc1f case_b #second number greater
[00400068] 46000306 mov.s $f12, $f0 ; 36: mov.s $f12, $f0
[0040006c] 34020002 ori $2, $0, 2 ; 37: li $v0, 2
[00400070] 0000000c syscall ; 38: syscall
[00400074] 3c011001 lui $1, 4097 [state] ; 39: la $a0, state
[00400078] 34240040 ori $4, $1, 64 [state]
[0040007c] 34020004 ori $2, $0, 4 ; 40: li $v0, 4
[00400080] 0000000c syscall ; 41: syscall
[00400084] 46000b06 mov.s $f12, $f1 ; 42: mov.s $f12, $f1
[00400088] 34020002 ori $2, $0, 2 ; 43: li $v0, 2
[0040008c] 0000000c syscall ; 44: syscall
[00400090] 10000011 beq $0, $0, 68 [end-0x00400090]; 45: beq $zero, $zero, end
[00400094] 46000b06 mov.s $f12, $f1 ; 48: mov.s $f12, $f1
[00400098] 34020002 ori $2, $0, 2 ; 49: li $v0, 2
[0040009c] 0000000c syscall ; 50: syscall
[004000a0] 3c011001 lui $1, 4097 [state] ; 51: la $a0, state
[004000a4] 34240040 ori $4, $1, 64 [state]
[004000a8] 34020004 ori $2, $0, 4 ; 52: li $v0, 4
[004000ac] 0000000c syscall ; 53: syscall
[004000b0] 46000306 mov.s $f12, $f0 ; 54: mov.s $f12, $f0
[004000b4] 34020002 ori $2, $0, 2 ; 55: li $v0, 2
[004000b8] 0000000c syscall ; 56: syscall
[004000bc] 10000006 beq $0, $0, 24 [end-0x004000bc]; 57: beq $zero, $zero, end
[004000c0] 3c011001 lui $1, 4097 [equall] ; 61: la $a0, equall
[004000c4] 34240029 ori $4, $1, 41 [equall]
[004000c8] 34020004 ori $2, $0, 4 ; 62: li $v0, 4
[004000cc] 0000000c syscall ; 63: syscall
[004000d0] 10000001 beq $0, $0, 4 [end-0x004000d0]; 64: beq $zero, $zero, end
[004000d4] 3402000a ori $2, $0, 10 ; 68: li $v0, 10 # system call code f
[004000d8] 0000000c syscall ; 69: syscall # call operating system
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Console
Enter first number:54
Enter second number:-23.5
54.00000000 is greater than -23.50000000

```
#Date: 2026-09-12
#By: Divyam Sood
#Leap Year

.data
input: .asciiz "Enter year:"
yes: .asciiz "Input year is a leap year"
no: .asciiz "Input year is not a leap year"



.globl main
.text

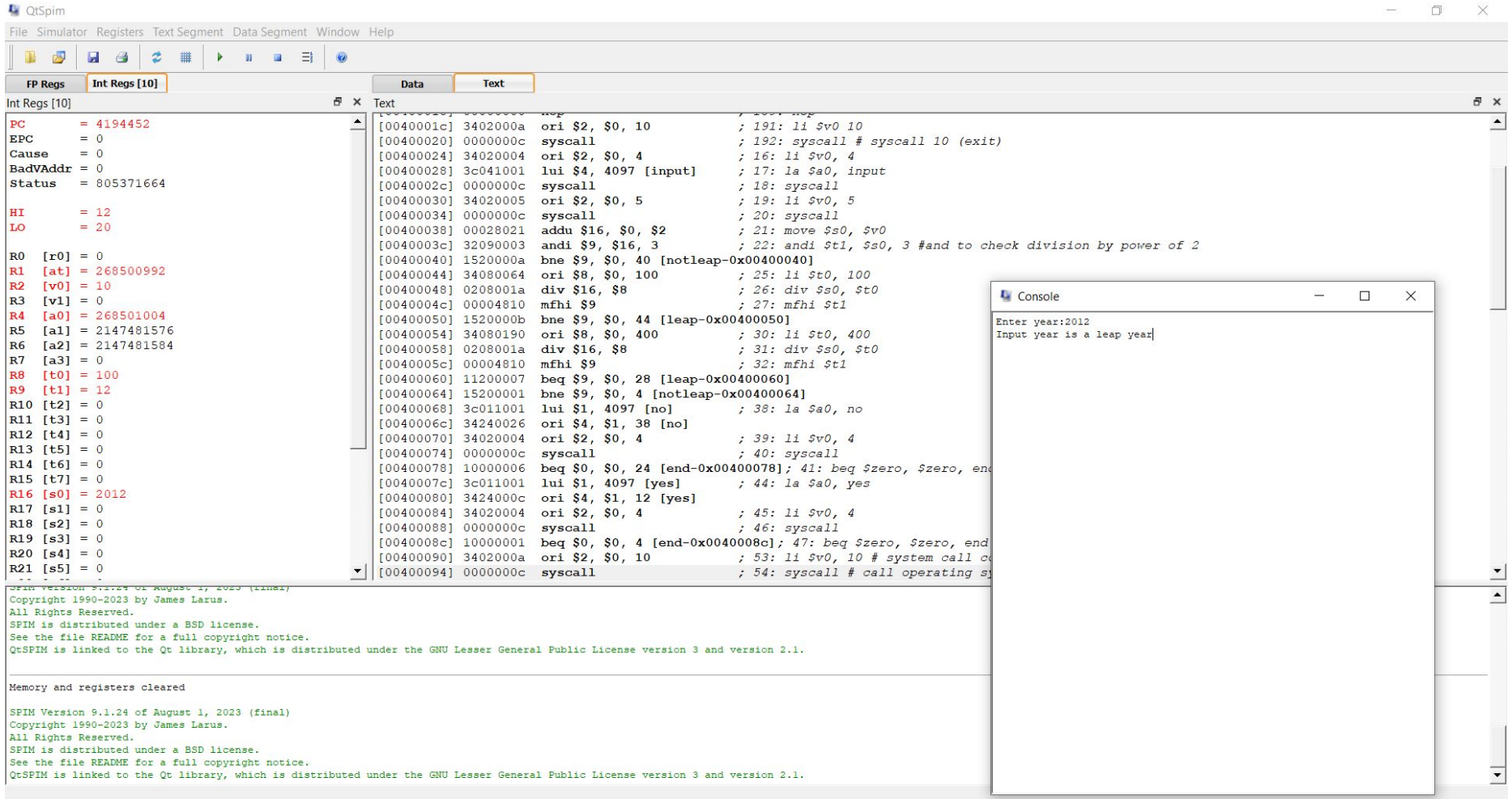
main:
    li $v0, 4
    la $a0, input
    syscall
    li $v0, 5
    syscall
    move $s0, $v0
    andi $t1, $s0, 3 #and to check division by power of 2
    bne $t1, $zero, notleap
    #This means year is divisible by 4. now we check divisible by 100.
    li $t0, 100
    div $s0, $t0
    mfhi $t1
    bne $t1, $zero, leap #divisible by 4 and not by 100
    #Now divisble by 4 and 100. so should be divisble by 400
    li $t0, 400
    div $s0, $t0
    mfhi $t1
    beq $t1, $zero, leap
    bne $t1, $zero, notleap

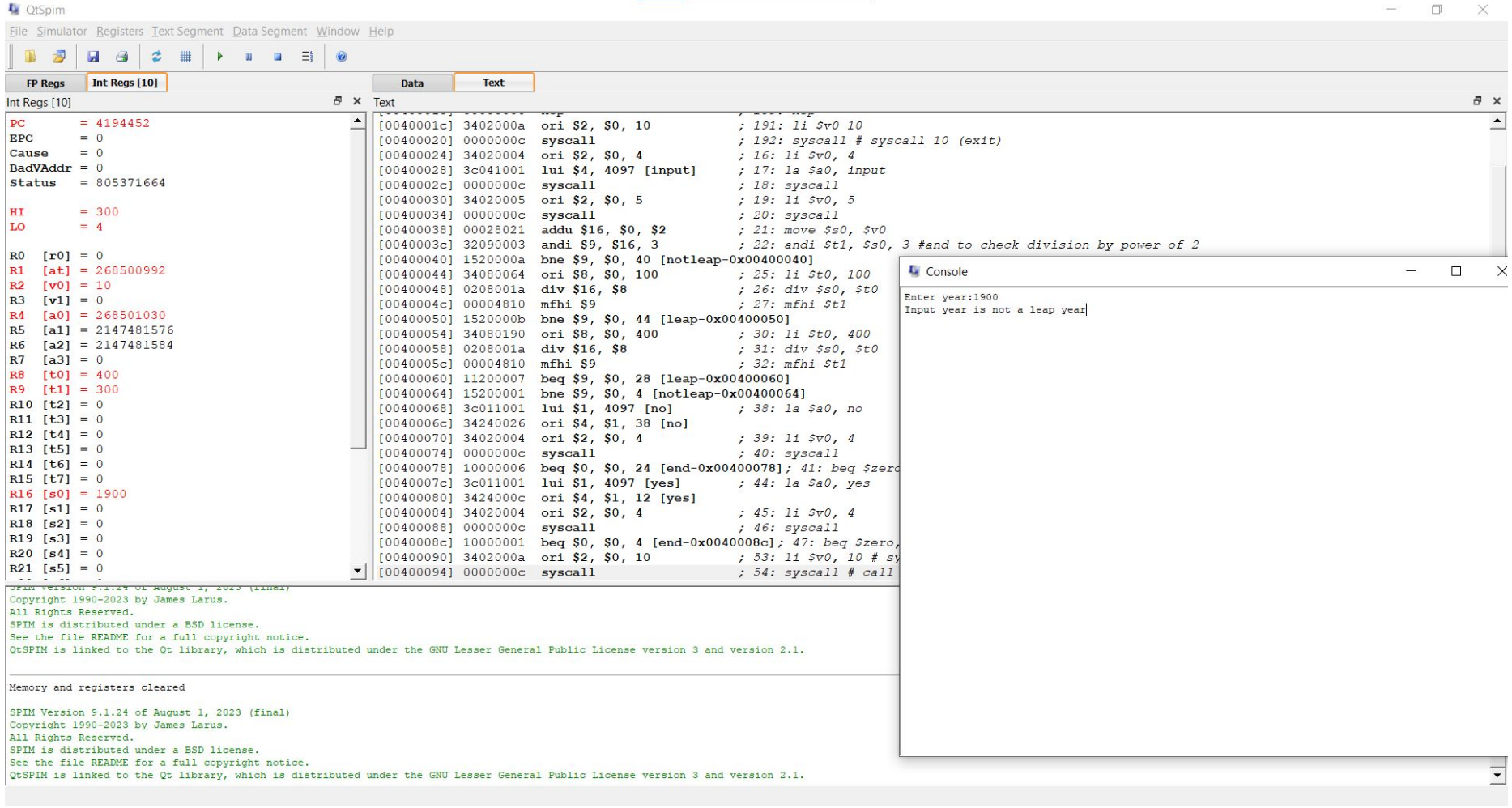

notleap:
    la $a0, no
    li $v0, 4
    syscall
    beq $zero, $zero, end

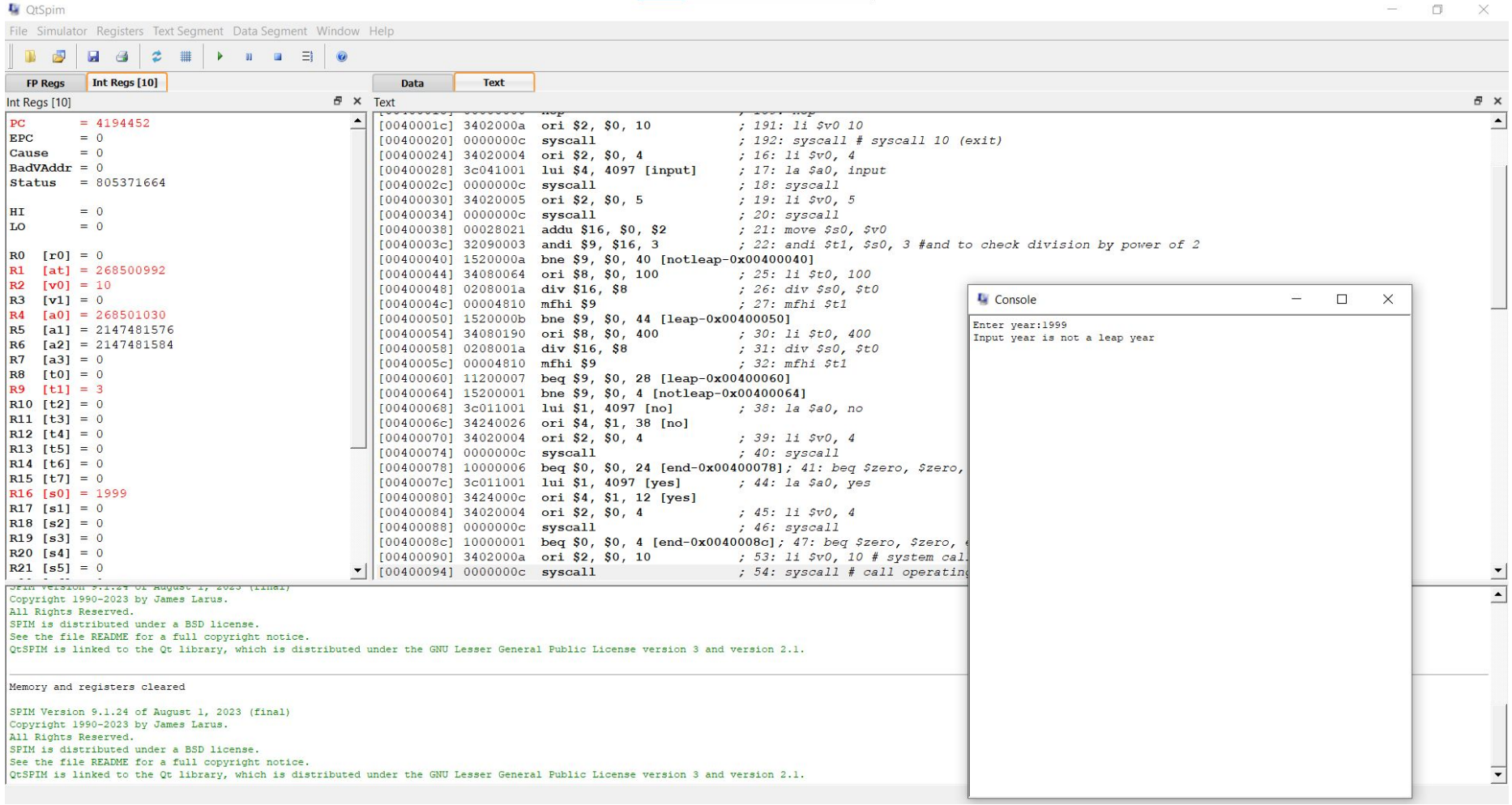
leap:
    la $a0, yes
    li $v0, 4
    syscall
    beq $zero, $zero, end

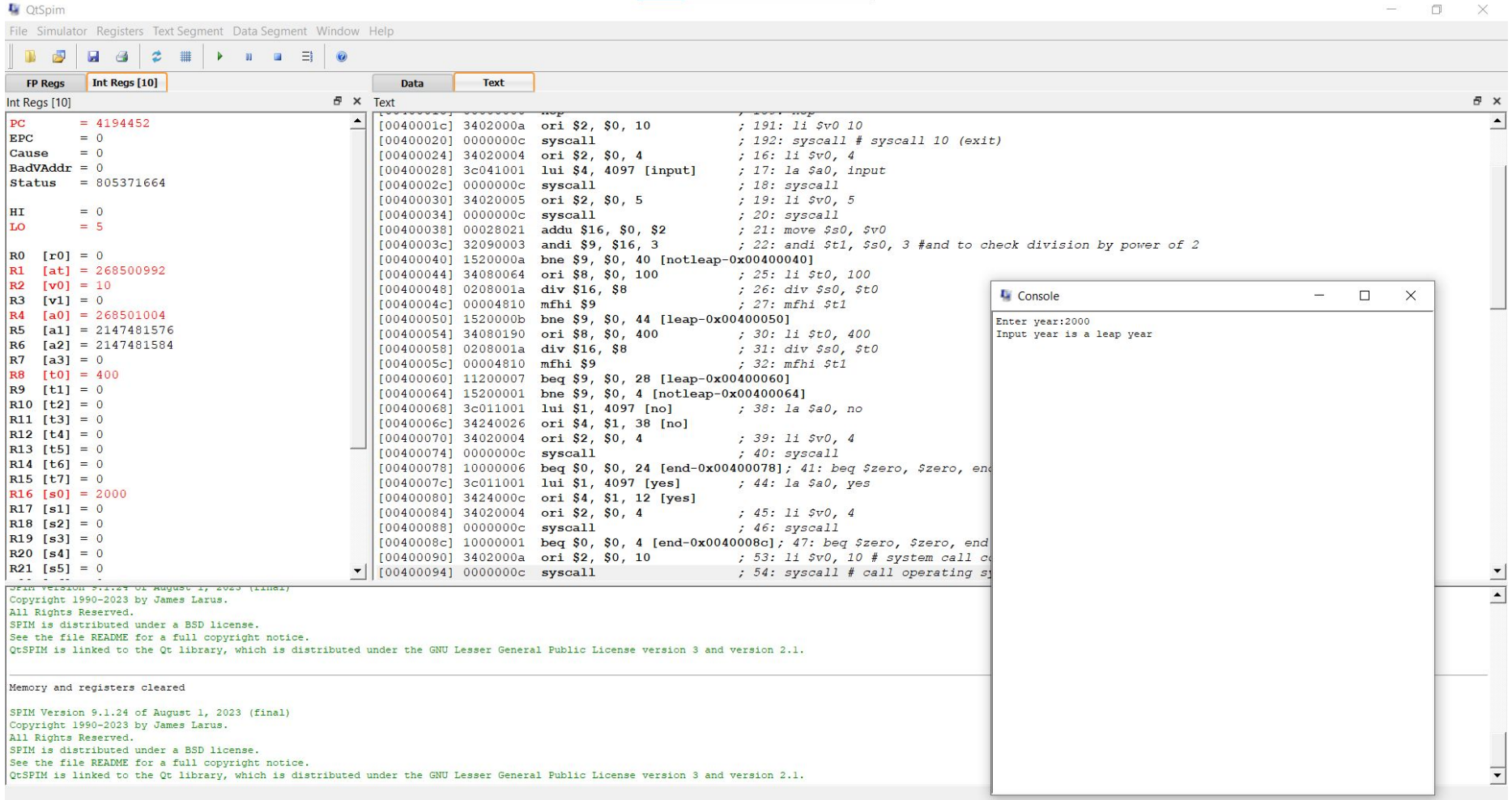


end:
    # exit program
    li  $v0, 10         # system call code for exit = 10
    syscall             # call operating system
```


QtSpim
File Simulator Registers Text Segment Data Segment Window Help
FP Regs
Int Regs [10]
Int Regs [10]
PC = 4194452
EPC = 0
Cause = 0
BadVAddr = 0
Status = 805371664
HI = 12
LO = 20
R0 [r0] = 0
R1 [at] = 268500992
R2 [v0] = 10
R3 [v1] = 0
R4 [a0] = 268501004
R5 [a1] = 2147481576
R6 [a2] = 2147481584
R7 [a3] = 0
R8 [t0] = 100
R9 [t1] = 12
R10 [t2] = 0
R11 [t3] = 0
R12 [t4] = 0
R13 [t5] = 0
R14 [t6] = 0
R15 [t7] = 0
R16 [s0] = 2012
R17 [s1] = 0
R18 [s2] = 0
R19 [s3] = 0
R20 [s4] = 0
R21 [s5] = 0
Data
Text
Text
[0040001c] 3402000a ori $2, $0, 10 ; 191: li $v0 10
[00400020] 0000000c syscall ; 192: syscall # syscall 10 (exit)
[00400024] 34020004 ori $2, $0, 4 ; 16: li $v0, 4
[00400028] 3c041001 lui $4, 4097 [input] ; 17: la $a0, input
[0040002c] 0000000c syscall ; 18: syscall
[00400030] 34020005 ori $2, $0, 5 ; 19: li $v0, 5
[00400034] 0000000c syscall ; 20: syscall
[00400038] 00028021 addu $16, $0, $2 ; 21: move $s0, $v0
[0040003c] 32090003 andi $9, $16, 3 ; 22: andi $t1, $s0, 3 #and to check division by power of 2
[00400040] 1520000a bne $9, $0, 40 [notleap-0x00400040]
[00400044] 34080064 ori $8, $0, 100 ; 25: li $t0, 100
[00400048] 0208001a div $16, $8 ; 26: div $s0, $t0
[0040004c] 00004810 mfhi $9 ; 27: mfhi $t1
[00400050] 1520000b bne $9, $0, 44 [leap-0x00400050]
[00400054] 34080190 ori $8, $0, 400 ; 30: li $t0, 400
[00400058] 0208001a div $16, $8 ; 31: div $s0, $t0
[0040005c] 00004810 mfhi $9 ; 32: mfhi $t1
[00400060] 11200007 beq $9, $0, 28 [leap-0x00400060]
[00400064] 15200001 bne $9, $0, 4 [notleap-0x00400064]
[00400068] 3c011001 lui $1, 4097 [no] ; 38: la $a0, no
[0040006c] 34240026 ori $4, $1, 38 [no]
[00400070] 34020004 ori $2, $0, 4 ; 39: li $v0, 4
[00400074] 0000000c syscall ; 40: syscall
[00400078] 10000006 beq $0, $0, 24 [end-0x00400078] ; 41: beq $zero, $zero, en
[0040007c] 3c011001 lui $1, 4097 [yes] ; 44: la $a0, yes
[00400080] 3424000c ori $4, $1, 12 [yes]
[00400084] 34020004 ori $2, $0, 4 ; 45: li $v0, 4
[00400088] 0000000c syscall ; 46: syscall
[0040008c] 10000001 beq $0, $0, 4 [end-0x0040008c] ; 47: beq $zero, $zero, end
[00400090] 3402000a ori $2, $0, 10 ; 53: li $v0, 10 # system call c
[00400094] 0000000c syscall ; 54: syscall # call operating s
Console
Enter year:2012
Input year is a leap year
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QtSpim
File Simulator Registers Text Segment Data Segment Window Help
FP Regs
Int Regs [10]
Int Regs [10]
PC = 4194452
EPC = 0
Cause = 0
BadVAddr = 0
Status = 805371664
HI = 300
LO = 4
R0 [r0] = 0
R1 [at] = 268500992
R2 [v0] = 10
R3 [v1] = 0
R4 [a0] = 268501030
R5 [a1] = 2147481576
R6 [a2] = 2147481584
R7 [a3] = 0
R8 [t0] = 400
R9 [t1] = 300
R10 [t2] = 0
R11 [t3] = 0
R12 [t4] = 0
R13 [t5] = 0
R14 [t6] = 0
R15 [t7] = 0
R16 [s0] = 1900
R17 [s1] = 0
R18 [s2] = 0
R19 [s3] = 0
R20 [s4] = 0
R21 [s5] = 0
Data
Text
Text
[0040001c] 3402000a ori $2, $0, 10 ; 191: li $v0 10
[00400020] 0000000c syscall ; 192: syscall # syscall 10 (exit)
[00400024] 34020004 ori $2, $0, 4 ; 16: li $v0, 4
[00400028] 3c041001 lui $4, 4097 [input] ; 17: la $a0, input
[0040002c] 0000000c syscall ; 18: syscall
[00400030] 34020005 ori $2, $0, 5 ; 19: li $v0, 5
[00400034] 0000000c syscall ; 20: syscall
[00400038] 00028021 addu $16, $0, $2 ; 21: move $s0, $v0
[0040003c] 32090003 andi $9, $16, 3 ; 22: andi $t1, $s0, 3 #and to check division by power of 2
[00400040] 1520000a bne $9, $0, 40 [notleap-0x00400040]
[00400044] 34080064 ori $8, $0, 100 ; 25: li $t0, 100
[00400048] 0208001a div $16, $8 ; 26: div $s0, $t0
[0040004c] 00004810 mfhi $9 ; 27: mfhi $t1
[00400050] 1520000b bne $9, $0, 44 [leap-0x00400050]
[00400054] 34080190 ori $8, $0, 400 ; 30: li $t0, 400
[00400058] 0208001a div $16, $8 ; 31: div $s0, $t0
[0040005c] 00004810 mfhi $9 ; 32: mfhi $t1
[00400060] 11200007 beq $9, $0, 28 [leap-0x00400060]
[00400064] 15200001 bne $9, $0, 4 [notleap-0x00400064]
[00400068] 3c011001 lui $1, 4097 [no] ; 38: la $a0, no
[0040006c] 34240026 ori $4, $1, 38 [no]
[00400070] 34020004 ori $2, $0, 4 ; 39: li $v0, 4
[00400074] 0000000c syscall ; 40: syscall
[00400078] 10000006 beq $0, $0, 24 [end-0x00400078] ; 41: beq $zero
[0040007c] 3c011001 lui $1, 4097 [yes] ; 44: la $a0, yes
[00400080] 3424000c ori $4, $1, 12 [yes]
[00400084] 34020004 ori $2, $0, 4 ; 45: li $v0, 4
[00400088] 0000000c syscall ; 46: syscall
[0040008c] 10000001 beq $0, $0, 4 [end-0x0040008c] ; 47: beq $zero,
[00400090] 3402000a ori $2, $0, 10 ; 53: li $v0, 10 # sy
[00400094] 0000000c syscall ; 54: syscall # call
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Console
Enter year:1900
Input year is not a leap year


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File Simulator Registers Text Segment Data Segment Window Help
FP Regs
Int Regs [10]
Int Regs [10]
PC = 4194452
EPC = 0
Cause = 0
BadVAddr = 0
Status = 805371664
HI = 0
LO = 0
R0 [r0] = 0
R1 [at] = 268500992
R2 [v0] = 10
R3 [v1] = 0
R4 [a0] = 268501030
R5 [a1] = 2147481576
R6 [a2] = 2147481584
R7 [a3] = 0
R8 [t0] = 0
R9 [t1] = 3
R10 [t2] = 0
R11 [t3] = 0
R12 [t4] = 0
R13 [t5] = 0
R14 [t6] = 0
R15 [t7] = 0
R16 [s0] = 1999
R17 [s1] = 0
R18 [s2] = 0
R19 [s3] = 0
R20 [s4] = 0
R21 [s5] = 0
Data
Text
Text
[0040001c] 3402000a ori $2, $0, 10 ; 191: li $v0 10
[00400020] 0000000c syscall ; 192: syscall # syscall 10 (exit)
[00400024] 34020004 ori $2, $0, 4 ; 16: li $v0, 4
[00400028] 3c041001 lui $4, 4097 [input] ; 17: la $a0, input
[0040002c] 0000000c syscall ; 18: syscall
[00400030] 34020005 ori $2, $0, 5 ; 19: li $v0, 5
[00400034] 0000000c syscall ; 20: syscall
[00400038] 00028021 addu $16, $0, $2 ; 21: move $s0, $v0
[0040003c] 32090003 andi $9, $16, 3 ; 22: andi $t1, $s0, 3 #and to check division by power of 2
[00400040] 1520000a bne $9, $0, 40 [notleap-0x00400040]
[00400044] 34080064 ori $8, $0, 100 ; 25: li $t0, 100
[00400048] 0208001a div $16, $8 ; 26: div $s0, $t0
[0040004c] 00004810 mfhi $9 ; 27: mfhi $t1
[00400050] 1520000b bne $9, $0, 44 [leap-0x00400050]
[00400054] 34080190 ori $8, $0, 400 ; 30: li $t0, 400
[00400058] 0208001a div $16, $8 ; 31: div $s0, $t0
[0040005c] 00004810 mfhi $9 ; 32: mfhi $t1
[00400060] 11200007 beq $9, $0, 28 [leap-0x00400060]
[00400064] 15200001 bne $9, $0, 4 [notleap-0x00400064]
[00400068] 3c011001 lui $1, 4097 [no] ; 38: la $a0, no
[0040006c] 34240026 ori $4, $1, 38 [no]
[00400070] 34020004 ori $2, $0, 4 ; 39: li $v0, 4
[00400074] 0000000c syscall ; 40: syscall
[00400078] 10000006 beq $0, $0, 24 [end-0x00400078] ; 41: beq $zero, $zero,
[0040007c] 3c011001 lui $1, 4097 [yes] ; 44: la $a0, yes
[00400080] 3424000c ori $4, $1, 12 [yes]
[00400084] 34020004 ori $2, $0, 4 ; 45: li $v0, 4
[00400088] 0000000c syscall ; 46: syscall
[0040008c] 10000001 beq $0, $0, 4 [end-0x0040008c] ; 47: beq $zero, $zero,
[00400090] 3402000a ori $2, $0, 10 ; 53: li $v0, 10 # system cal
[00400094] 0000000c syscall ; 54: syscall # call operatin
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Memory and registers cleared
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Console
Enter year:1999
Input year is not a leap year


QtSpim
File Simulator Registers Text Segment Data Segment Window Help
FP Regs
Int Regs [10]
Int Regs [10]
PC = 4194452
EPC = 0
Cause = 0
BadVAddr = 0
Status = 805371664
HI = 0
LO = 5
R0 [r0] = 0
R1 [at] = 268500992
R2 [v0] = 10
R3 [v1] = 0
R4 [a0] = 268501004
R5 [a1] = 2147481576
R6 [a2] = 2147481584
R7 [a3] = 0
R8 [t0] = 400
R9 [t1] = 0
R10 [t2] = 0
R11 [t3] = 0
R12 [t4] = 0
R13 [t5] = 0
R14 [t6] = 0
R15 [t7] = 0
R16 [s0] = 2000
R17 [s1] = 0
R18 [s2] = 0
R19 [s3] = 0
R20 [s4] = 0
R21 [s5] = 0
Data
Text
Text
[0040001c] 3402000a ori $2, $0, 10 ; 191: li $v0 10
[00400020] 0000000c syscall ; 192: syscall # syscall 10 (exit)
[00400024] 34020004 ori $2, $0, 4 ; 16: li $v0, 4
[00400028] 3c041001 lui $4, 4097 [input] ; 17: la $a0, input
[0040002c] 0000000c syscall ; 18: syscall
[00400030] 34020005 ori $2, $0, 5 ; 19: li $v0, 5
[00400034] 0000000c syscall ; 20: syscall
[00400038] 00028021 addu $16, $0, $2 ; 21: move $s0, $v0
[0040003c] 32090003 andi $9, $16, 3 ; 22: andi $t1, $s0, 3 #and to check division by power of 2
[00400040] 1520000a bne $9, $0, 40 [notleap-0x00400040]
[00400044] 34080064 ori $8, $0, 100 ; 25: li $t0, 100
[00400048] 0208001a div $16, $8 ; 26: div $s0, $t0
[0040004c] 00004810 mfhi $9 ; 27: mfhi $t1
[00400050] 1520000b bne $9, $0, 44 [leap-0x00400050]
[00400054] 34080190 ori $8, $0, 400 ; 30: li $t0, 400
[00400058] 0208001a div $16, $8 ; 31: div $s0, $t0
[0040005c] 00004810 mfhi $9 ; 32: mfhi $t1
[00400060] 11200007 beq $9, $0, 28 [leap-0x00400060]
[00400064] 15200001 bne $9, $0, 4 [notleap-0x00400064]
[00400068] 3c011001 lui $1, 4097 [no] ; 38: la $a0, no
[0040006c] 34240026 ori $4, $1, 38 [no]
[00400070] 34020004 ori $2, $0, 4 ; 39: li $v0, 4
[00400074] 0000000c syscall ; 40: syscall
[00400078] 10000006 beq $0, $0, 24 [end-0x00400078] ; 41: beq $zero, $zero, en
[0040007c] 3c011001 lui $1, 4097 [yes] ; 44: la $a0, yes
[00400080] 3424000c ori $4, $1, 12 [yes]
[00400084] 34020004 ori $2, $0, 4 ; 45: li $v0, 4
[00400088] 0000000c syscall ; 46: syscall
[0040008c] 10000001 beq $0, $0, 4 [end-0x0040008c] ; 47: beq $zero, $zero, end
[00400090] 3402000a ori $2, $0, 10 ; 53: li $v0, 10 # system call c
[00400094] 0000000c syscall ; 54: syscall # call operating s
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Console
Enter year:2000
Input year is a leap year

```
#Date: 2026-09-11
#By: Divyam Sood
#Exam Eligibility

.data
name: .space 20
tot: .word 0
num: .word 0
p1: .asciiz "Enter your name please:"
p2: .asciiz "Enter total number of classes:"
p3: .asciiz "Enter number of classes you attended:"
sorry: .asciiz "You cannot attend more classes than total classes!"
pass: .asciiz " is allowed to sit in the exam"
fail: .asciiz " is not allowed to sit in the exam"
newline: .asciiz "\n"

.globl main
.text

main:
    #Get player name
    li $v0, 4
    la $a0, p1
    syscall
    li $a1, 21
    la $a0, name
    li $v0, 8
    syscall


    #now to remove the enter character (ASCII 10) from input
    la $t7, name
    li $s1, 10

remove:
    addi $t7, $t7, 1
    lb $t6, 0($t7)
    bne $t6, $s1, remove
    beq $zero, $zero, continue


continue:
    sb $zero, 0($t7) #change the \n character
    #Get total classes
    li $v0, 4
    la $a0, p2
    syscall
    li $v0, 5
    syscall
    move $t0, $v0
    #$t0 now has total classes

    #Get attended classes
    li $v0, 4
    la $a0, p3
    syscall
    li $v0, 5
    syscall
    move $t1, $v0
    #$t1 now has attended classes

    bgt $t1, $t0, fool #Case of total classes less than attended

    li $t3, 100
    mul $t1, $t3, $t1
    div $t1, $t0
    mflo $t2 #Quotient
    mfhi $t4 #Remainder
    li $t6, 75
```

```
    bge $t2, $t6, yes
    blt $t2, $t6, no

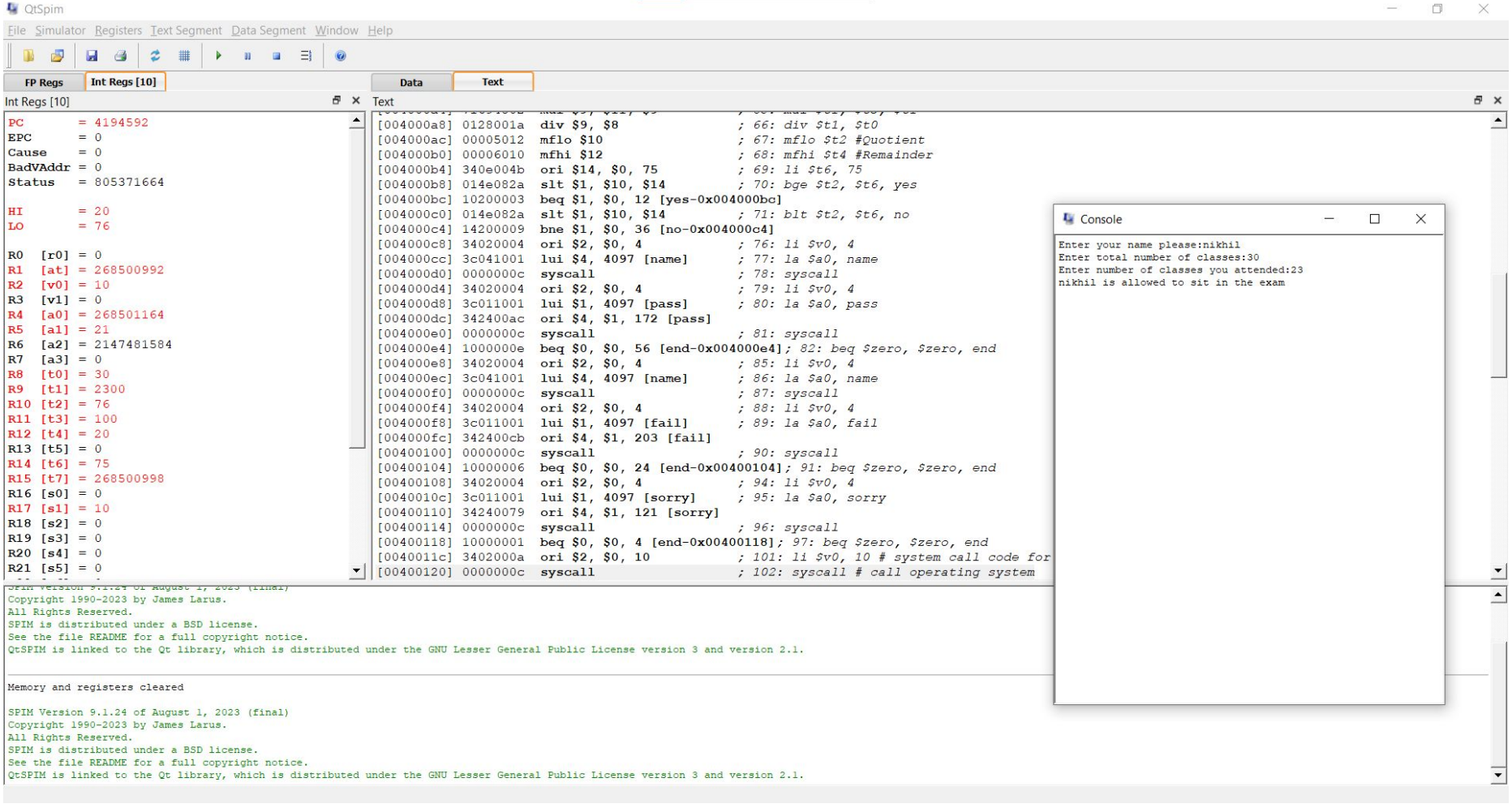


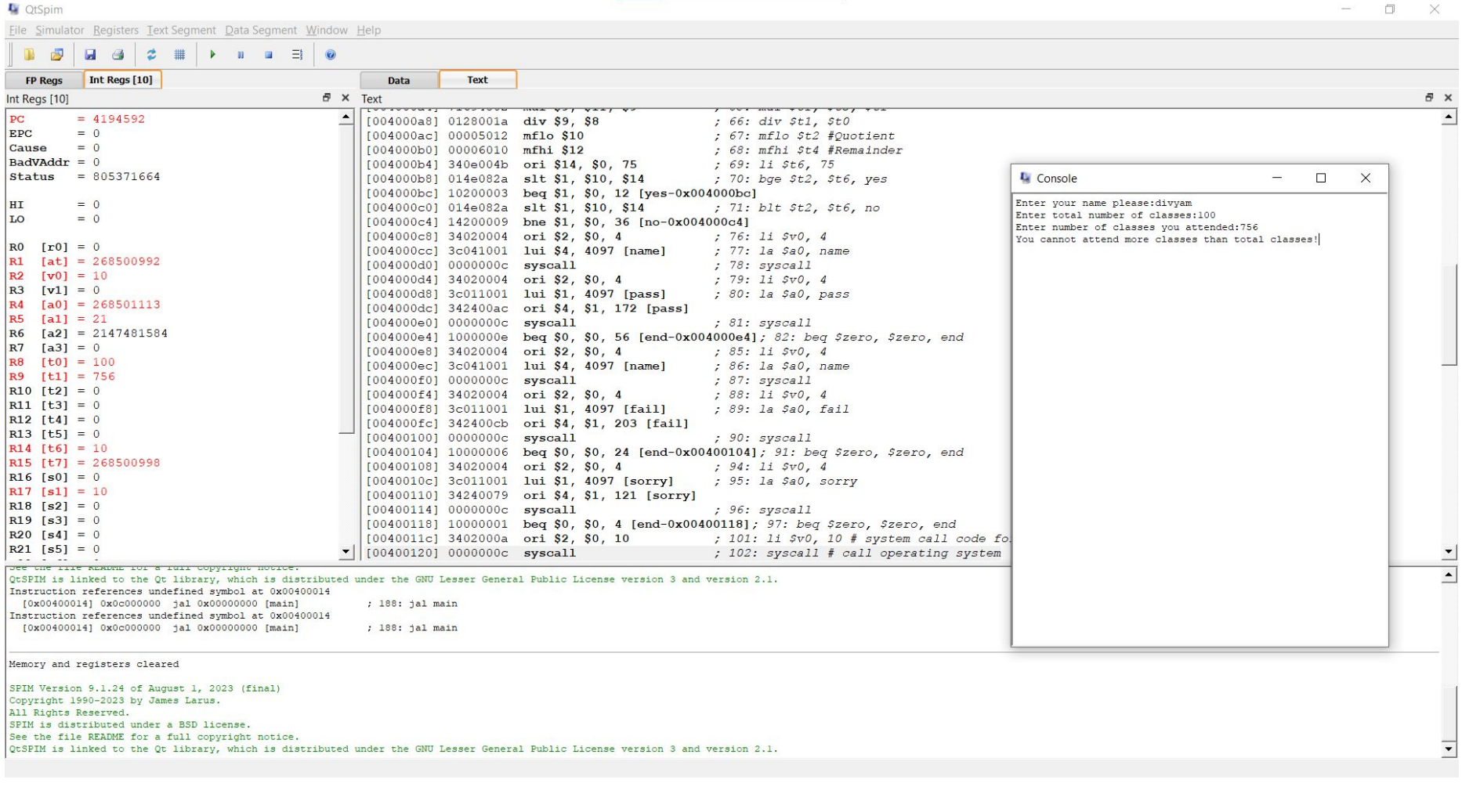
yes:
    li $v0, 4
    la $a0, name
    syscall
    li $v0, 4
    la $a0, pass
    syscall
    beq $zero, $zero, end

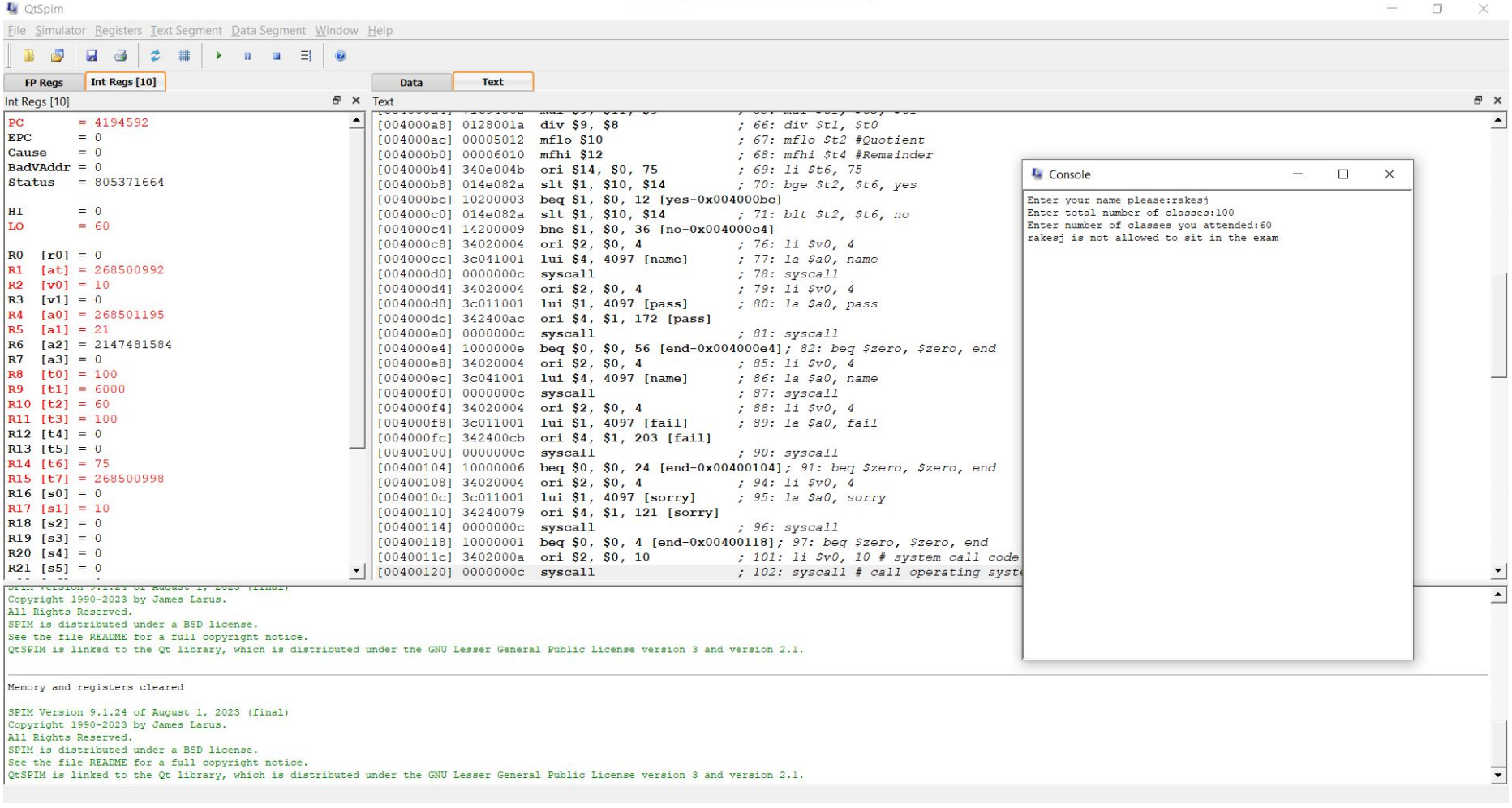
no:
    li $v0, 4
    la $a0, name
    syscall
    li $v0, 4
    la $a0, fail
    syscall
    beq $zero, $zero, end

fool:
    li $v0, 4
    la $a0, sorry
    syscall
    beq $zero, $zero, end

end:
    # exit program
    li  $v0, 10         # system call code for exit = 10
    syscall             # call operating system
```


QtSpim
File Simulator Registers Text Segment Data Segment Window Help
FP Regs
Int Regs [10]
Int Regs [10]
PC = 4194592
EPC = 0
Cause = 0
BadVAddr = 0
Status = 805371664
HI = 20
LO = 76
R0 [r0] = 0
R1 [at] = 268500992
R2 [v0] = 10
R3 [v1] = 0
R4 [a0] = 268501164
R5 [a1] = 21
R6 [a2] = 2147481584
R7 [a3] = 0
R8 [t0] = 30
R9 [t1] = 2300
R10 [t2] = 76
R11 [t3] = 100
R12 [t4] = 20
R13 [t5] = 0
R14 [t6] = 75
R15 [t7] = 268500998
R16 [s0] = 0
R17 [s1] = 10
R18 [s2] = 0
R19 [s3] = 0
R20 [s4] = 0
R21 [s5] = 0
Data
Text
Text
[004000a8] 0128001a div $9, $8 ; 66: div $t1, $t0
[004000ac] 00005012 mflo $10 ; 67: mflo $t2 #Quotient
[004000b0] 00006010 mfhi $12 ; 68: mfhi $t4 #Remainder
[004000b4] 340e004b ori $14, $0, 75 ; 69: li $t6, 75
[004000b8] 014e082a slt $1, $10, $14 ; 70: bge $t2, $t6, yes
[004000bc] 10200003 beq $1, $0, 12 [yes-0x004000bc]
[004000c0] 014e082a slt $1, $10, $14 ; 71: blt $t2, $t6, no
[004000c4] 14200009 bne $1, $0, 36 [no-0x004000c4]
[004000c8] 34020004 ori $2, $0, 4 ; 76: li $v0, 4
[004000cc] 3c041001 lui $4, 4097 [name] ; 77: la $a0, name
[004000d0] 0000000c syscall ; 78: syscall
[004000d4] 34020004 ori $2, $0, 4 ; 79: li $v0, 4
[004000d8] 3c011001 lui $1, 4097 [pass] ; 80: la $a0, pass
[004000dc] 342400ac ori $4, $1, 172 [pass]
[004000e0] 0000000c syscall ; 81: syscall
[004000e4] 1000000e beq $0, $0, 56 [end-0x004000e4] ; 82: beq $zero, $zero, end
[004000e8] 34020004 ori $2, $0, 4 ; 85: li $v0, 4
[004000ec] 3c041001 lui $4, 4097 [name] ; 86: la $a0, name
[004000f0] 0000000c syscall ; 87: syscall
[004000f4] 34020004 ori $2, $0, 4 ; 88: li $v0, 4
[004000f8] 3c011001 lui $1, 4097 [fail] ; 89: la $a0, fail
[004000fc] 342400cb ori $4, $1, 203 [fail]
[00400100] 0000000c syscall ; 90: syscall
[00400104] 10000006 beq $0, $0, 24 [end-0x00400104] ; 91: beq $zero, $zero, end
[00400108] 34020004 ori $2, $0, 4 ; 94: li $v0, 4
[0040010c] 3c011001 lui $1, 4097 [sorry] ; 95: la $a0, sorry
[00400110] 34240079 ori $4, $1, 121 [sorry]
[00400114] 0000000c syscall ; 96: syscall
[00400118] 10000001 beq $0, $0, 4 [end-0x00400118] ; 97: beq $zero, $zero, end
[0040011c] 3402000a ori $2, $0, 10 ; 101: li $v0, 10 # system call code for
[00400120] 0000000c syscall ; 102: syscall # call operating system
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Console
Enter your name please:nikhil
Enter total number of classes:30
Enter number of classes you attended:23
nikhil is allowed to sit in the exam


QtSpim
File Simulator Registers Text Segment Data Segment Window Help
FP Regs
Int Regs [10]
Int Regs [10]
PC = 4194592
EPC = 0
Cause = 0
BadVAddr = 0
Status = 805371664
HI = 0
LO = 0
R0 [r0] = 0
R1 [at] = 268500992
R2 [v0] = 10
R3 [v1] = 0
R4 [a0] = 268501113
R5 [a1] = 21
R6 [a2] = 2147481584
R7 [a3] = 0
R8 [t0] = 100
R9 [t1] = 756
R10 [t2] = 0
R11 [t3] = 0
R12 [t4] = 0
R13 [t5] = 0
R14 [t6] = 10
R15 [t7] = 268500998
R16 [s0] = 0
R17 [s1] = 10
R18 [s2] = 0
R19 [s3] = 0
R20 [s4] = 0
R21 [s5] = 0
Data
Text
Text
[004000a8] 0128001a div $9, $8 ; 66: div $t1, $t0
[004000ac] 00005012 mflo $10 ; 67: mflo $t2 #Quotient
[004000b0] 00006010 mfhi $12 ; 68: mfhi $t4 #Remainder
[004000b4] 340e004b ori $14, $0, 75 ; 69: li $t6, 75
[004000b8] 014e082a slt $1, $10, $14 ; 70: bge $t2, $t6, yes
[004000bc] 10200003 beq $1, $0, 12 [yes-0x004000bc]
[004000c0] 014e082a slt $1, $10, $14 ; 71: blt $t2, $t6, no
[004000c4] 14200009 bne $1, $0, 36 [no-0x004000c4]
[004000c8] 34020004 ori $2, $0, 4 ; 76: li $v0, 4
[004000cc] 3c041001 lui $4, 4097 [name] ; 77: la $a0, name
[004000d0] 0000000c syscall ; 78: syscall
[004000d4] 34020004 ori $2, $0, 4 ; 79: li $v0, 4
[004000d8] 3c011001 lui $1, 4097 [pass] ; 80: la $a0, pass
[004000dc] 342400ac ori $4, $1, 172 [pass]
[004000e0] 0000000c syscall ; 81: syscall
[004000e4] 1000000e beq $0, $0, 56 [end-0x004000e4] ; 82: beq $zero, $zero, end
[004000e8] 34020004 ori $2, $0, 4 ; 85: li $v0, 4
[004000ec] 3c041001 lui $4, 4097 [name] ; 86: la $a0, name
[004000f0] 0000000c syscall ; 87: syscall
[004000f4] 34020004 ori $2, $0, 4 ; 88: li $v0, 4
[004000f8] 3c011001 lui $1, 4097 [fail] ; 89: la $a0, fail
[004000fc] 342400cb ori $4, $1, 203 [fail]
[00400100] 0000000c syscall ; 90: syscall
[00400104] 10000006 beq $0, $0, 24 [end-0x00400104] ; 91: beq $zero, $zero, end
[00400108] 34020004 ori $2, $0, 4 ; 94: li $v0, 4
[0040010c] 3c011001 lui $1, 4097 [sorry] ; 95: la $a0, sorry
[00400110] 34240079 ori $4, $1, 121 [sorry]
[00400114] 0000000c syscall ; 96: syscall
[00400118] 10000001 beq $0, $0, 4 [end-0x00400118] ; 97: beq $zero, $zero, end
[0040011c] 3402000a ori $2, $0, 10 ; 101: li $v0, 10 # system call code fo
[00400120] 0000000c syscall ; 102: syscall # call operating system
Console
Enter your name please:divyam
Enter total number of classes:100
Enter number of classes you attended:756
You cannot attend more classes than total classes!
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Instruction references undefined symbol at 0x00400014
[0x00400014] 0x0c000000 jal 0x00000000 [main] ; 188: jal main
Instruction references undefined symbol at 0x00400014
[0x00400014] 0x0c000000 jal 0x00000000 [main] ; 188: jal main
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File Simulator Registers Text Segment Data Segment Window Help
FP Regs
Int Regs [10]
Int Regs [10]
PC = 4194592
EPC = 0
Cause = 0
BadVAddr = 0
Status = 805371664
HI = 0
LO = 60
R0 [r0] = 0
R1 [at] = 268500992
R2 [v0] = 10
R3 [v1] = 0
R4 [a0] = 268501195
R5 [a1] = 21
R6 [a2] = 2147481584
R7 [a3] = 0
R8 [t0] = 100
R9 [t1] = 6000
R10 [t2] = 60
R11 [t3] = 100
R12 [t4] = 0
R13 [t5] = 0
R14 [t6] = 75
R15 [t7] = 268500998
R16 [s0] = 0
R17 [s1] = 10
R18 [s2] = 0
R19 [s3] = 0
R20 [s4] = 0
R21 [s5] = 0
Data
Text
Text
[004000a8] 0128001a div $9, $8 ; 66: div $t1, $t0
[004000ac] 00005012 mflo $10 ; 67: mflo $t2 #Quotient
[004000b0] 00006010 mfhi $12 ; 68: mfhi $t4 #Remainder
[004000b4] 340e004b ori $14, $0, 75 ; 69: li $t6, 75
[004000b8] 014e082a slt $1, $10, $14 ; 70: bge $t2, $t6, yes
[004000bc] 10200003 beq $1, $0, 12 [yes-0x004000bc]
[004000c0] 014e082a slt $1, $10, $14 ; 71: blt $t2, $t6, no
[004000c4] 14200009 bne $1, $0, 36 [no-0x004000c4]
[004000c8] 34020004 ori $2, $0, 4 ; 76: li $v0, 4
[004000cc] 3c041001 lui $4, 4097 [name] ; 77: la $a0, name
[004000d0] 0000000c syscall ; 78: syscall
[004000d4] 34020004 ori $2, $0, 4 ; 79: li $v0, 4
[004000d8] 3c011001 lui $1, 4097 [pass] ; 80: la $a0, pass
[004000dc] 342400ac ori $4, $1, 172 [pass]
[004000e0] 0000000c syscall ; 81: syscall
[004000e4] 1000000e beq $0, $0, 56 [end-0x004000e4] ; 82: beq $zero, $zero, end
[004000e8] 34020004 ori $2, $0, 4 ; 85: li $v0, 4
[004000ec] 3c041001 lui $4, 4097 [name] ; 86: la $a0, name
[004000f0] 0000000c syscall ; 87: syscall
[004000f4] 34020004 ori $2, $0, 4 ; 88: li $v0, 4
[004000f8] 3c011001 lui $1, 4097 [fail] ; 89: la $a0, fail
[004000fc] 342400cb ori $4, $1, 203 [fail]
[00400100] 0000000c syscall ; 90: syscall
[00400104] 10000006 beq $0, $0, 24 [end-0x00400104] ; 91: beq $zero, $zero, end
[00400108] 34020004 ori $2, $0, 4 ; 94: li $v0, 4
[0040010c] 3c011001 lui $1, 4097 [sorry] ; 95: la $a0, sorry
[00400110] 34240079 ori $4, $1, 121 [sorry]
[00400114] 0000000c syscall ; 96: syscall
[00400118] 10000001 beq $0, $0, 4 [end-0x00400118] ; 97: beq $zero, $zero, end
[0040011c] 3402000a ori $2, $0, 10 ; 101: li $v0, 10 # system call code
[00400120] 0000000c syscall ; 102: syscall # call operating syst
Console
Enter your name please:rakesj
Enter total number of classes:100
Enter number of classes you attended:60
rakesj is not allowed to sit in the exam
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```
#Date: 2026-09-12
#By: Divyam Sood
#Marks Range

.data
input: .asciiz "Enter your marks (out of 100):"
a: .asciiz "Grade A assigned"
b_b: .asciiz "Grade B assigned"
c: .asciiz "Grade C assigned"
f: .asciiz "Grade F assigned"

.globl main
.text

main:
    li $v0, 4
    la $a0, input
    syscall
    li $v0, 6
    syscall
    li.s $f2, 80.0
    c.lt.s $f0, $f2
    bc1f grade_a
    #Here means score less than 80
    li.s $f2, 60.0
    c.lt.s $f0, $f2
    bc1f grade_b
    #Here means score less than 60
    li.s $f2, 40.0
    c.lt.s $f0, $f2
    bc1f grade_c
    #Here means score less than 40
    beq $zero, $zero, grade_f



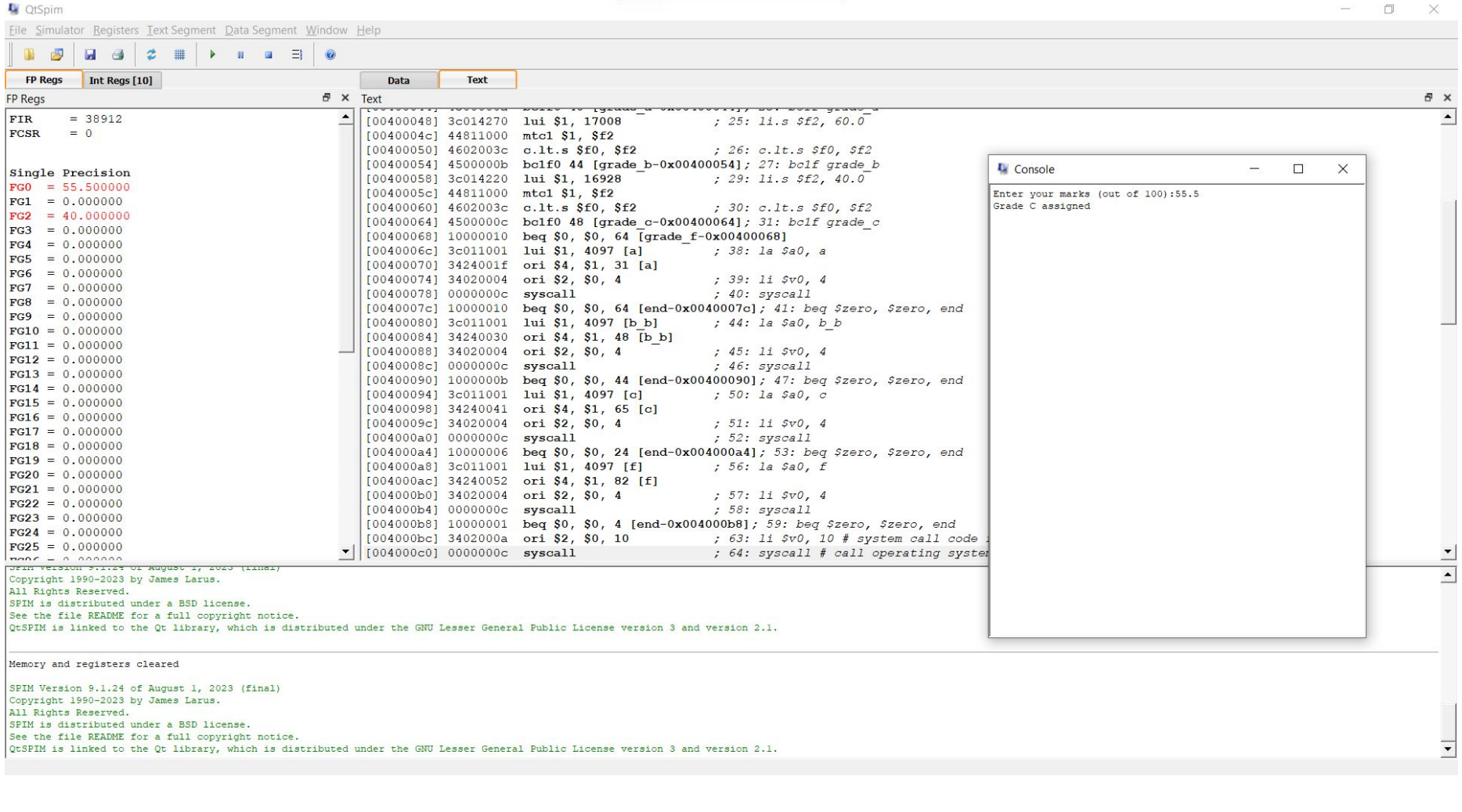
grade_a:
    la $a0, a
    li $v0, 4
    syscall
    beq $zero, $zero, end

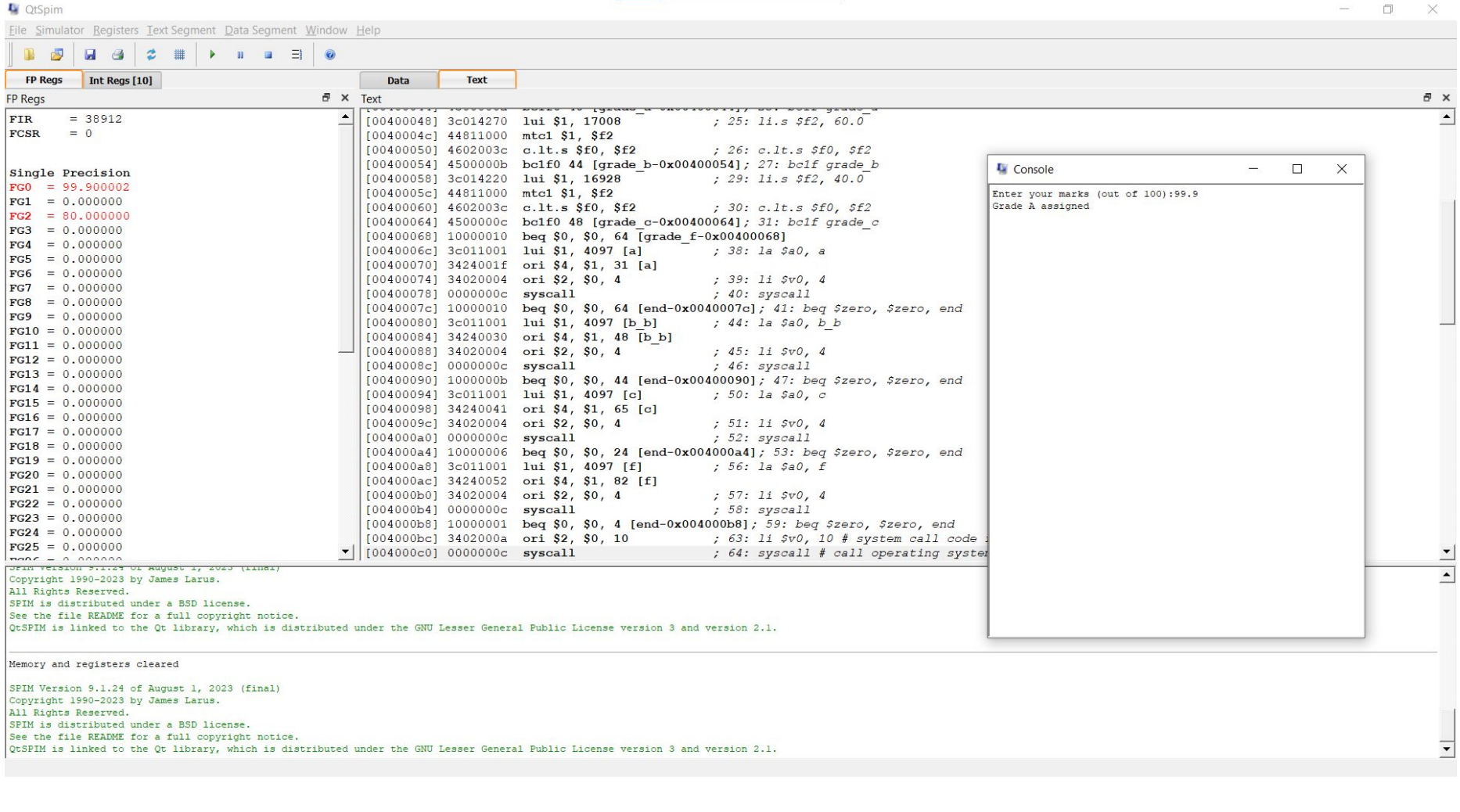
grade_b:
    la $a0, b_b
    li $v0, 4
    syscall
    beq $zero, $zero, end

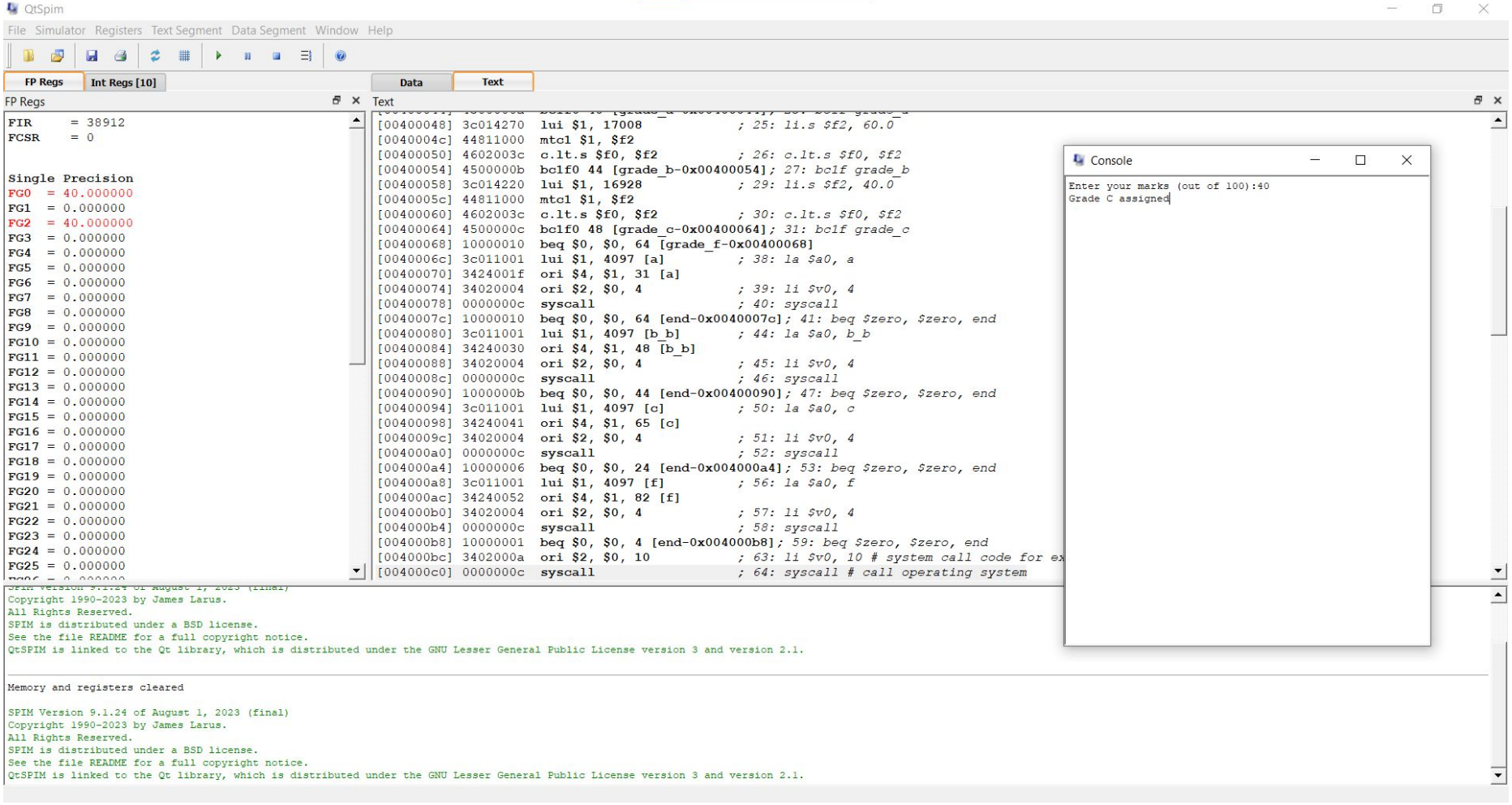
grade_c:
    la $a0, c
    li $v0, 4
    syscall
    beq $zero, $zero, end

grade_f:
    la $a0, f
    li $v0, 4
    syscall
    beq $zero, $zero, end

end:
    # exit program
    li  $v0, 10         # system call code for exit = 10
    syscall             # call operating system
```


QtSpim
File Simulator Registers Text Segment Data Segment Window Help
FP Regs
Int Regs [10]
Data
Text
FP Regs
FIR = 38912
FCSR = 0
Single Precision
FG0 = 55.500000
FG1 = 0.000000
FG2 = 40.000000
FG3 = 0.000000
FG4 = 0.000000
FG5 = 0.000000
FG6 = 0.000000
FG7 = 0.000000
FG8 = 0.000000
FG9 = 0.000000
FG10 = 0.000000
FG11 = 0.000000
FG12 = 0.000000
FG13 = 0.000000
FG14 = 0.000000
FG15 = 0.000000
FG16 = 0.000000
FG17 = 0.000000
FG18 = 0.000000
FG19 = 0.000000
FG20 = 0.000000
FG21 = 0.000000
FG22 = 0.000000
FG23 = 0.000000
FG24 = 0.000000
FG25 = 0.000000
Text
[00400048] 3c014270 lui $1, 17008 ; 25: li.s $f2, 60.0
[0040004c] 44811000 mtc1 $1, $f2
[00400050] 4602003c c.lt.s $f0, $f2 ; 26: c.lt.s $f0, $f2
[00400054] 4500000b bc1f0 44 [grade_b-0x00400054] ; 27: bc1f grade_b
[00400058] 3c014220 lui $1, 16928 ; 29: li.s $f2, 40.0
[0040005c] 44811000 mtc1 $1, $f2
[00400060] 4602003c c.lt.s $f0, $f2 ; 30: c.lt.s $f0, $f2
[00400064] 4500000c bc1f0 48 [grade_c-0x00400064] ; 31: bc1f grade_c
[00400068] 10000010 beq $0, $0, 64 [grade_f-0x00400068]
[0040006c] 3c011001 lui $1, 4097 [a] ; 38: la $a0, a
[00400070] 3424001f ori $4, $1, 31 [a]
[00400074] 34020004 ori $2, $0, 4 ; 39: li $v0, 4
[00400078] 0000000c syscall ; 40: syscall
[0040007c] 10000010 beq $0, $0, 64 [end-0x0040007c] ; 41: beq $zero, $zero, end
[00400080] 3c011001 lui $1, 4097 [b_b] ; 44: la $a0, b_b
[00400084] 34240030 ori $4, $1, 48 [b_b]
[00400088] 34020004 ori $2, $0, 4 ; 45: li $v0, 4
[0040008c] 0000000c syscall ; 46: syscall
[00400090] 1000000b beq $0, $0, 44 [end-0x00400090] ; 47: beq $zero, $zero, end
[00400094] 3c011001 lui $1, 4097 [c] ; 50: la $a0, c
[00400098] 34240041 ori $4, $1, 65 [c]
[0040009c] 34020004 ori $2, $0, 4 ; 51: li $v0, 4
[004000a0] 0000000c syscall ; 52: syscall
[004000a4] 10000006 beq $0, $0, 24 [end-0x004000a4] ; 53: beq $zero, $zero, end
[004000a8] 3c011001 lui $1, 4097 [f] ; 56: la $a0, f
[004000ac] 34240052 ori $4, $1, 82 [f]
[004000b0] 34020004 ori $2, $0, 4 ; 57: li $v0, 4
[004000b4] 0000000c syscall ; 58: syscall
[004000b8] 10000001 beq $0, $0, 4 [end-0x004000b8] ; 59: beq $zero, $zero, end
[004000bc] 3402000a ori $2, $0, 10 ; 63: li $v0, 10 # system call code
[004000c0] 0000000c syscall ; 64: syscall # call operating syste
Console
Enter your marks (out of 100):55.5
Grade C assigned
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QtSpim
File Simulator Registers Text Segment Data Segment Window Help
FP Regs
Int Regs [10]
Data
Text
FP Regs
FIR = 38912
FCSR = 0
Single Precision
FG0 = 99.900002
FG1 = 0.000000
FG2 = 80.000000
FG3 = 0.000000
FG4 = 0.000000
FG5 = 0.000000
FG6 = 0.000000
FG7 = 0.000000
FG8 = 0.000000
FG9 = 0.000000
FG10 = 0.000000
FG11 = 0.000000
FG12 = 0.000000
FG13 = 0.000000
FG14 = 0.000000
FG15 = 0.000000
FG16 = 0.000000
FG17 = 0.000000
FG18 = 0.000000
FG19 = 0.000000
FG20 = 0.000000
FG21 = 0.000000
FG22 = 0.000000
FG23 = 0.000000
FG24 = 0.000000
FG25 = 0.000000
Text
[00400048] 3c014270 lui $1, 17008 ; 25: li.s $f2, 60.0
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[00400060] 4602003c c.lt.s $f0, $f2 ; 30: c.lt.s $f0, $f2
[00400064] 4500000c bc1f0 48 [grade_c-0x00400064] ; 31: bc1f grade_c
[00400068] 10000010 beq $0, $0, 64 [grade_f-0x00400068]
[0040006c] 3c011001 lui $1, 4097 [a] ; 38: la $a0, a
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[0040007c] 10000010 beq $0, $0, 64 [end-0x0040007c] ; 41: beq $zero, $zero, end
[00400080] 3c011001 lui $1, 4097 [b_b] ; 44: la $a0, b_b
[00400084] 34240030 ori $4, $1, 48 [b_b]
[00400088] 34020004 ori $2, $0, 4 ; 45: li $v0, 4
[0040008c] 0000000c syscall ; 46: syscall
[00400090] 1000000b beq $0, $0, 44 [end-0x00400090] ; 47: beq $zero, $zero, end
[00400094] 3c011001 lui $1, 4097 [c] ; 50: la $a0, c
[00400098] 34240041 ori $4, $1, 65 [c]
[0040009c] 34020004 ori $2, $0, 4 ; 51: li $v0, 4
[004000a0] 0000000c syscall ; 52: syscall
[004000a4] 10000006 beq $0, $0, 24 [end-0x004000a4] ; 53: beq $zero, $zero, end
[004000a8] 3c011001 lui $1, 4097 [f] ; 56: la $a0, f
[004000ac] 34240052 ori $4, $1, 82 [f]
[004000b0] 34020004 ori $2, $0, 4 ; 57: li $v0, 4
[004000b4] 0000000c syscall ; 58: syscall
[004000b8] 10000001 beq $0, $0, 4 [end-0x004000b8] ; 59: beq $zero, $zero, end
[004000bc] 3402000a ori $2, $0, 10 ; 63: li $v0, 10 # system call code
[004000c0] 0000000c syscall ; 64: syscall # call operating syste
Console
Enter your marks (out of 100):99.9
Grade A assigned
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File Simulator Registers Text Segment Data Segment Window Help
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Text
FP Regs
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Single Precision
FG0 = 40.000000
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FG2 = 40.000000
FG3 = 0.000000
FG4 = 0.000000
FG5 = 0.000000
FG6 = 0.000000
FG7 = 0.000000
FG8 = 0.000000
FG9 = 0.000000
FG10 = 0.000000
FG11 = 0.000000
FG12 = 0.000000
FG13 = 0.000000
FG14 = 0.000000
FG15 = 0.000000
FG16 = 0.000000
FG17 = 0.000000
FG18 = 0.000000
FG19 = 0.000000
FG20 = 0.000000
FG21 = 0.000000
FG22 = 0.000000
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[00400058] 3c014220 lui $1, 16928 ; 29: li.s $f2, 40.0
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[00400064] 4500000c bc1f0 48 [grade_c-0x00400064] ; 31: bc1f grade_c
[00400068] 10000010 beq $0, $0, 64 [grade_f-0x00400068]
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[00400074] 34020004 ori $2, $0, 4 ; 39: li $v0, 4
[00400078] 0000000c syscall ; 40: syscall
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[00400080] 3c011001 lui $1, 4097 [b_b] ; 44: la $a0, b_b
[00400084] 34240030 ori $4, $1, 48 [b_b]
[00400088] 34020004 ori $2, $0, 4 ; 45: li $v0, 4
[0040008c] 0000000c syscall ; 46: syscall
[00400090] 1000000b beq $0, $0, 44 [end-0x00400090] ; 47: beq $zero, $zero, end
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[004000a8] 3c011001 lui $1, 4097 [f] ; 56: la $a0, f
[004000ac] 34240052 ori $4, $1, 82 [f]
[004000b0] 34020004 ori $2, $0, 4 ; 57: li $v0, 4
[004000b4] 0000000c syscall ; 58: syscall
[004000b8] 10000001 beq $0, $0, 4 [end-0x004000b8] ; 59: beq $zero, $zero, end
[004000bc] 3402000a ori $2, $0, 10 ; 63: li $v0, 10 # system call code for ex
[004000c0] 0000000c syscall ; 64: syscall # call operating system
Console
Enter your marks (out of 100):40
Grade C assigned
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